Mini Analysis 3

- In our analysis we look at the data set with outliers removed (tube strikes) and focus only on the weekend

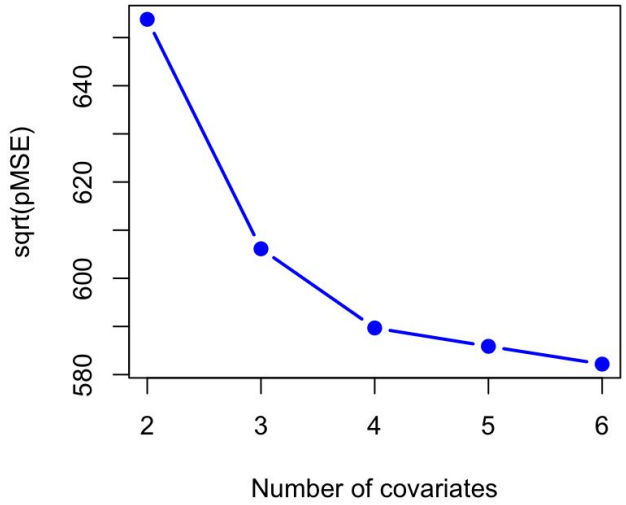
regsubsets analysis

- We divide the data into a training and test data set
- We look at regsubsets where we allow all continuous covariates and one categorical (seasons)
- season is forced in every model
- If we have a model with just season, then adding humidity first seems best

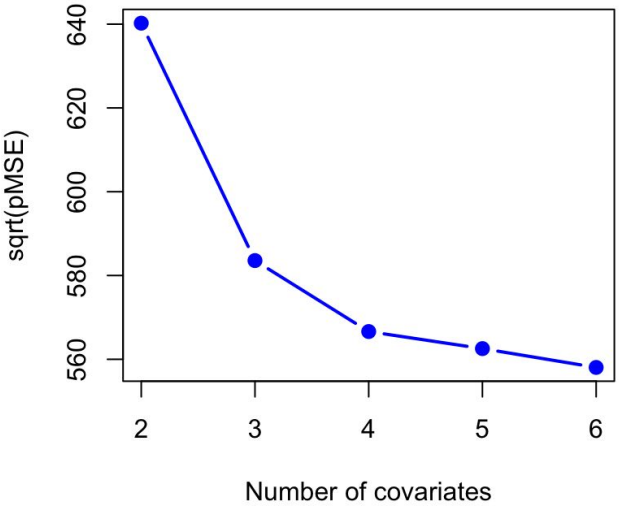
```
Selection Algorithm: exhaustive
          seasonAutumn seasonWinter seasonSpring t1  hrs hum wind_speed t2
4  ( 1 ) "*"          "*"          "*"          " " " " "*" " "        " "
5  ( 1 ) "*"          "*"          "*"          "*" " " "*" " "        " "
6  ( 1 ) "*"          "*"          "*"          "*" "*" "*" " "        " "
7  ( 1 ) "*"          "*"          "*"          "*" "*" "*" "*"        " "
8  ( 1 ) "*"          "*"          "*"          "*" "*" "*" "*"        "*"
```

regsubsets analysis - pMSE

70-30% training-test split



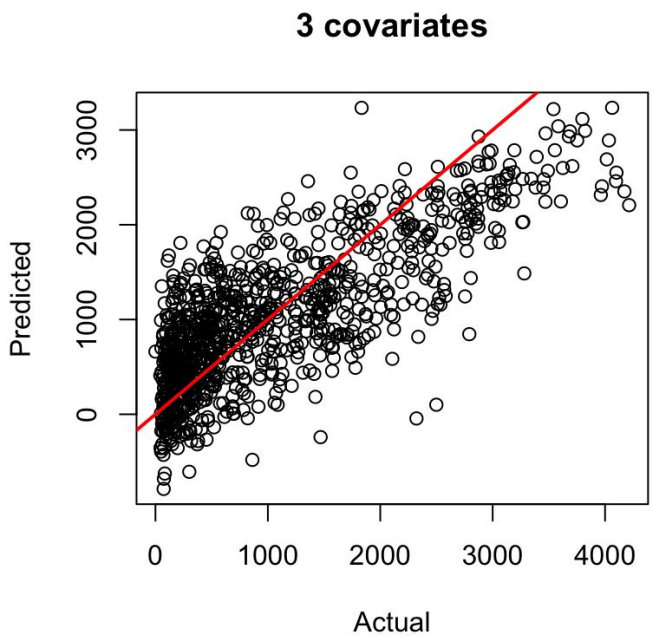
80-20% training-test split

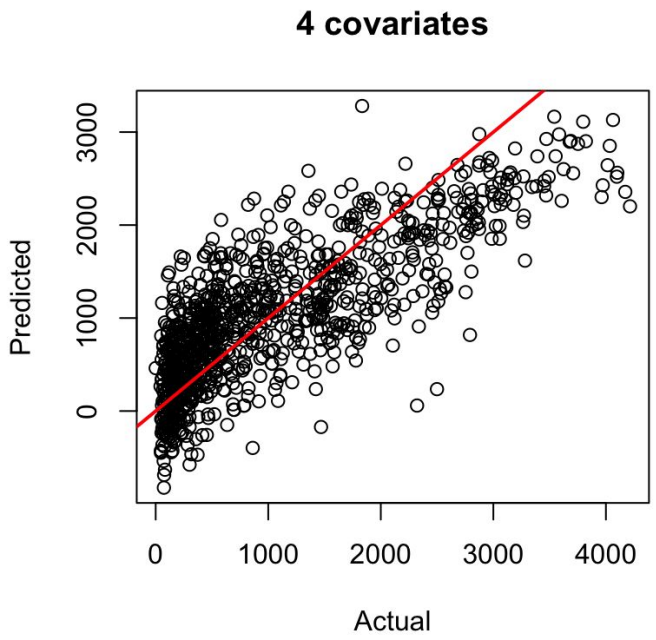


- Not a significant difference, both proportions of the training-test split seems to indicate that “3” or “4” covariates are the most fitting
- (1 categorical + 2 or 3 continuous)

Fitted responses vs actual test responses

- Doesn’t look quite linear, maybe a transformation would be beneficial



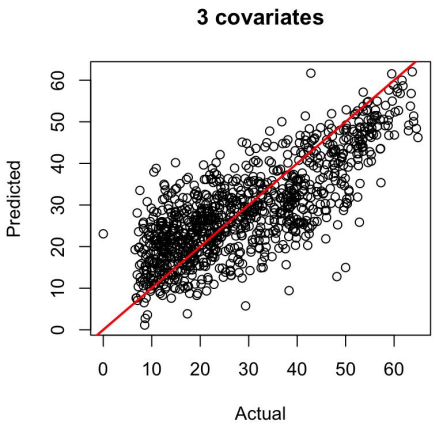


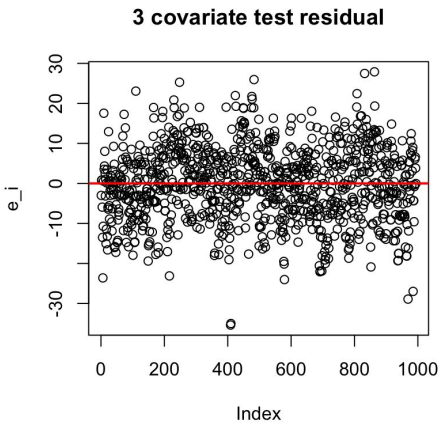
Generalization capability

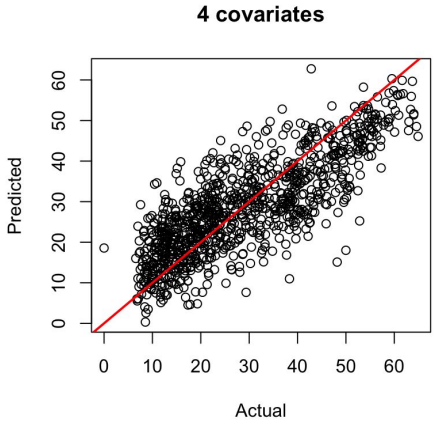
- We look at the models with 3 and 4 covariates. Using a sqrt transformation on cnt in the models and then computing the predictions on the test data we get the following
- Small changes in predictions, more apparent change in R^2

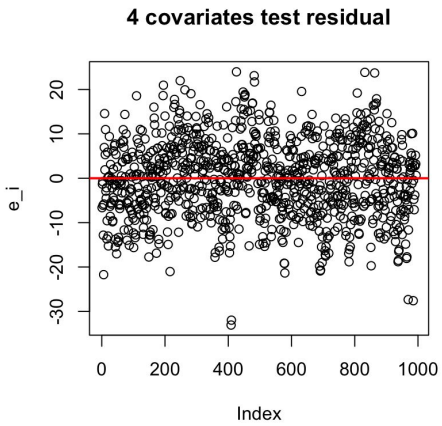
```
Multiple R-squared:  0.5662,
```

```
Multiple R-squared:  0.6155
```









K-fold cross validation

- As the previous results were quite dependent on the training and test split that was chosen, we want to use a cross validation to weaken the dependency on the selected data.
- We use K-fold as our data set is quite large
- Still, using the model with 4 covariates could still be justified as the average MSE over the 10 folds is quite large and the standard deviation from the mean decreases quite significantly between the folds

	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Fold 6	Fold 7	Fold 8	Fold 9	Fold 10	Average	SD
Model 1 (season + hum + t1)	80.4204	82.4452	88.8635	89.7868	79.6346	85.0283	87.6201	90.9233	92.9185	86.6751	86.4316	4.4857
Model 2 (season + hum + t1 + hrs)	74.6327	75.7723	77.3510	78.3417	72.9392	74.0070	79.7037	77.4244	81.9221	75.1234	76.7217	2.7678

Olsrr - Seven covariates, full data

Olsrr analysis applied to lm of seven covariates:

- t1
- t2
- hum
- wind speed
- hour of day
- season
- weather code

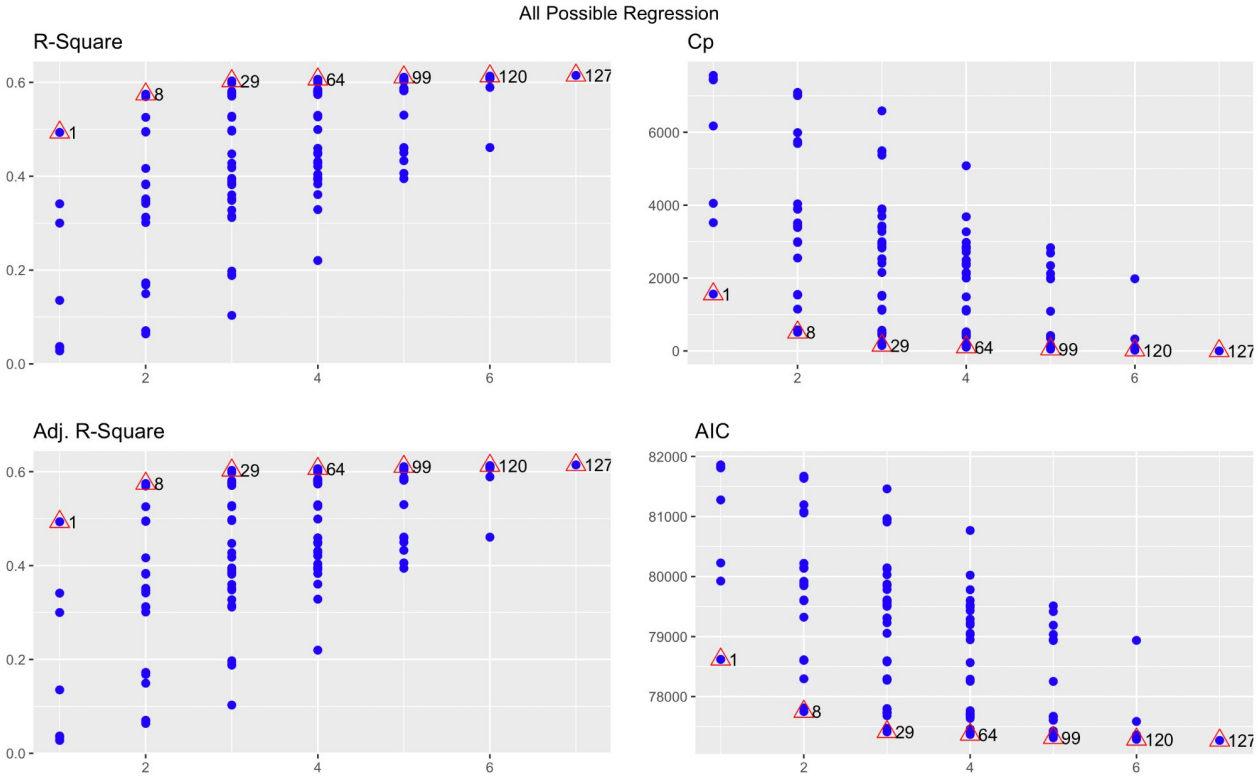
The “full” data applied was a subset which included all counts below 6000 and only weekends. Not divided into test and training data.

```
26  l1mod <- lm(cnt~t1+t2+hum+wind_speed+hour_of_day+season+weather_code,data=bike_data)
27  out <- ols_step_all_possible(l1mod)
28  plot(out)
```

Olsrr results - Full model

The results shows that all covariates together will create the “best” linear regression model.

However, applying 4 or 7 covariates in the model does not make much difference according to the graphs.

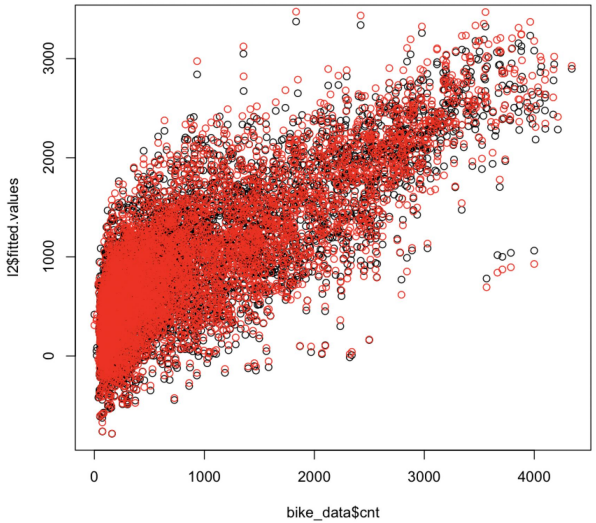


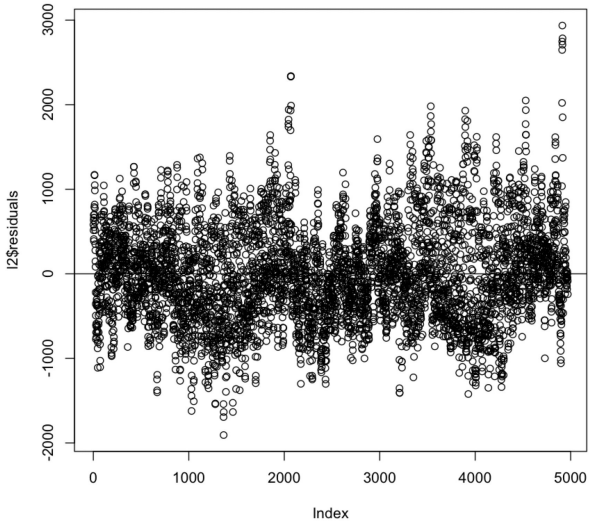
Olsrr results - Full model

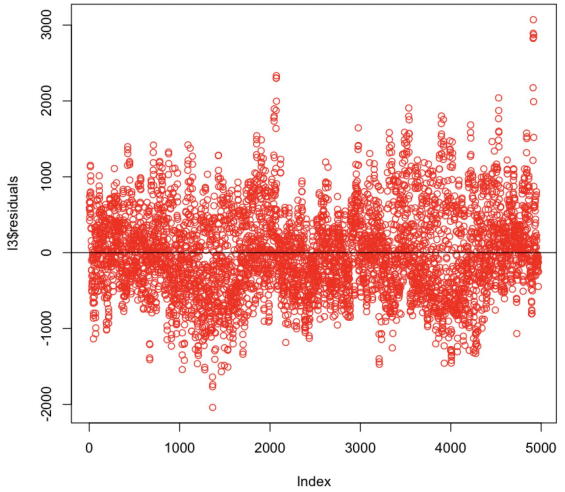
Looking at the best lm with 4 covariates vs all covariates:

(t1, hum, hour_of_day, season) gave the best lm when 4 covariates were used.

No real difference in fitted data between the two models.







Olsrr - Training vs Test data

Given that the difference in the fitted data was minimal, study the difference on test data by dividing data into training and test data.

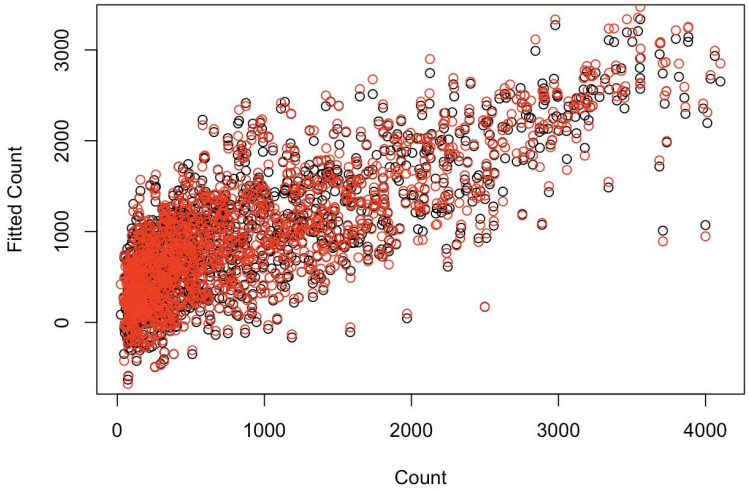
Training used to generate the model and test data used to predict “new” counts.

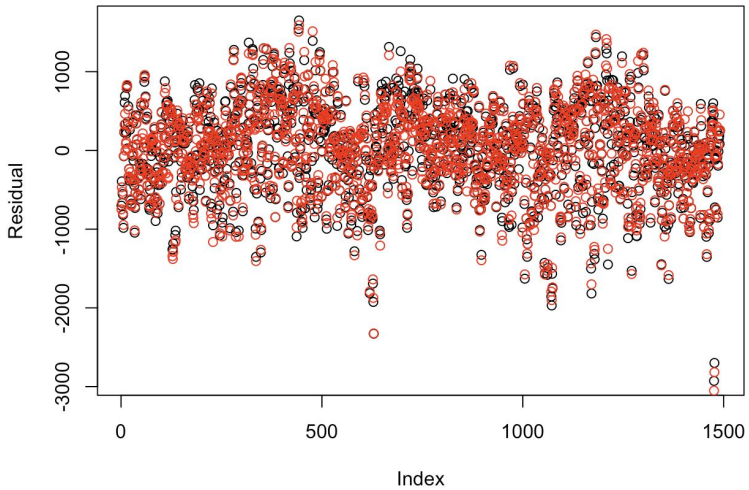
```
36  set.seed(123)
37  index <- sample(seq_len(nrow(bike_data)), size = 0.7 * nrow(bike_data))
38  train <- bike_data[index, ]
39  test  <- bike_data[-index, ]
```

```
41  trainl1mod <- lm(cnt~t1+t2+hum+wind_speed+hour_of_day+season+weather_code,data=train)
42  trainout <- ols_step_all_possible(trainl1mod)
```

Olsrr - Training vs Test data - Results

Both are “equally good” at predicting count from new data. Similar pattern and residuals between the models






```
127 cnt ~ t1 + t2 + hum + wind_speed + hour_of_day + season + weather_code 566.2194
    64                                 cnt ~ t1 + hum + hour_of_day + season 574.7086
```